

Quantum Graphs over 3×3 Matrices

Classification, Voltage Graphs, and Comparisons

Filip Rabięga, Dr. Mariusz Tobolski (supervisor)

September 11th, 2026

Roadmap

- C^* -algebras and crossed product C^* -algebras
- Quantum sets and quantum graphs
- Gromada's theorem: quantum graphs on $M_n(\mathbb{C}) \leftrightarrow$ subspaces
- Voltage and derived quantum graphs
- Quantum isomorphism

(brief recap)

- **Main content: quantum graphs on $M_3(\mathbb{C})$**
 - Classification via clock-shift matrices
 - Realization as classical derived graphs
 - Vertex-transitivity
 - Where the story breaks down

C^* -algebras and crossed product C^* -algebras

C^* -algebra: a Banach algebra A satisfying the C^* -axiom: $\|a^*a\| = \|a\|^2$ for all $a \in A$. I only consider unital C^* -algebras.

Crossed product C^* -algebra: a C^* -algebra $A \rtimes_\alpha G$ spanned by the functions $u_g : G \rightarrow A$,

$$u_g(h) = \begin{cases} 1_A, & h = g, \\ 0_A, & h \neq g, \end{cases} \quad h \in G,$$

where G is a finite group. Multiplication, involution and norm are determined by an action α of G on A .

Quantum sets and quantum graphs

Quantum set: a pair (B, ψ) – finite-dim. C^* -algebra B with a positive and faithful functional ψ such that $mm^* = \text{Id}$.

- Classical set $V \rightsquigarrow (\mathbb{C}^V, \psi_V)$
- $M_n(\mathbb{C}) \rightsquigarrow (M_n(\mathbb{C}), n \text{Tr})$ – a *genuinely noncommutative* example

Quantum graph: a linear map $A: B \rightarrow B$ generalizing an adjacency matrix, with

$$m(A \otimes A)m^* = A, \quad A(b^*) = A(b)^*.$$

- *Loop-free:* $m(A \otimes \text{Id})m^* = 0$
- *Undirected:* $A = A^*$

Gromada's theorem

Theorem (Gromada)

Quantum graphs on $M_n(\mathbb{C})$ with m quantum edges $\longleftrightarrow$ m -dimensional subspaces $V \subseteq \mathfrak{gl}_n$, via

$$A_V = n^2 \sum_{s=1}^m \xi_s \otimes \bar{\xi}_s.$$

- No loops $\iff V \subseteq \mathfrak{sl}(n) = \{A \in \mathfrak{gl}(n) : \text{Tr}(A) = 0\}$
- Undirected $\iff V = V^*$
- Isomorphic graphs $\iff V = U W U^*$, $U \in SU(n) = \{U \in U(n) : \det(U) = 1\}$

This turns questions about quantum graphs into **linear algebra**.

Voltage and derived quantum graphs

Classical picture (Gross–Tucker): label edges of a multigraph Γ by a group $M_3(\mathbb{C})$
 $\rightsquigarrow$ unfold into a derived graph Γ^λ on $V \times G$.

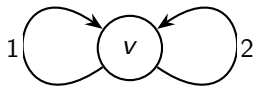
Quantum generalization: a $M_3(\mathbb{C})$ -indexed family $\tilde{A} = (\tilde{A}_g)_{g \in G}$ of quantum adjacency matrices, reassembled via a crossed product

$$\tilde{A} \rtimes_{\hat{\alpha}} \hat{G} := \sum_{g \in G} m(X_g \otimes E_{\rtimes}^* \tilde{A}_g E_{\rtimes}) m^*.$$

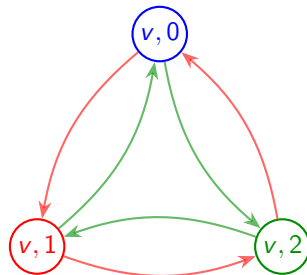
Key result (Schäfer–Tobolski)

Derived quantum graphs w.r.t. *different* actions on the same voltage data are always **quantum isomorphic**.

Example of a voltage and a derived graph



Base voltage graph Γ over $\mathbb{Z}_3$



Derived graph Γ^λ (two 3-cycles)

Figure: Voltage graph and derived cover over $\mathbb{Z}_3$, for a base graph with a single vertex and two loops.

Quantum isomorphism

Two quantum graphs A_1, A_2 are **quantum isomorphic**, $A_1 \cong_q A_2$, if there is a unitary $\mathcal{H}$ -twisted map ρ intertwining them.

- Taking $\mathcal{H} = \mathbb{C}$ recovers ordinary isomorphism
- $\cong_q$ is **strictly weaker** than $\cong$
- Originates in nonlocal games: quantum strategies can win games classical strategies cannot

This is the tool that lets us relate “truly quantum” graphs on $M_3(\mathbb{C})$ to *classical* graphs.

Generalised Pauli matrices on $M_3(\mathbb{C})$

Clock and shift matrices:

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \omega = e^{2\pi i/3}.$$

- $P^3 = Q^3 = I$, $PQ = \omega QP$
- $\{P^a Q^b\}_{a,b \in \{0,1,2\}}$ is an *orthogonal basis* of $M_3(\mathbb{C})$
- Higher-dimensional analogue of the Pauli matrices

Classifying subspaces generated by P, Q

Reflection-invariant blocks: $\mathcal{P}_1 = \{P, P^2\}$, $\mathcal{P}_2 = \{Q, Q^2\}$, $\mathcal{P}_3, \mathcal{P}_4$ (mixed).

m	subspaces	count
0	$V_0 = \{0\}$	1
2	$V_k = \text{span}(\mathcal{P}_k)$	4
4	$W_{i,j} = \text{span}(\mathcal{P}_i \oplus \mathcal{P}_j)$	6
6	$U_k = \text{span}(\cup_{i \neq k} \mathcal{P}_i)$	4
8	$\mathfrak{sl}_3$	1

16 self-reflective subspaces total, built from 2^4 block choices.

Full unitary equivalence

The discrete Fourier transform

$$F = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}$$

is not enough.

Key trick

A second unitary S (a phase gate) cyclically permutes $\mathcal{P}_2, \mathcal{P}_3, \mathcal{P}_4$. Together, F and S generate the full alternating group A_4

Corollary

All 16 subspaces collapse to exactly **5** isomorphism classes – one per dimension $m \in \{0, 2, 4, 6, 8\}$.

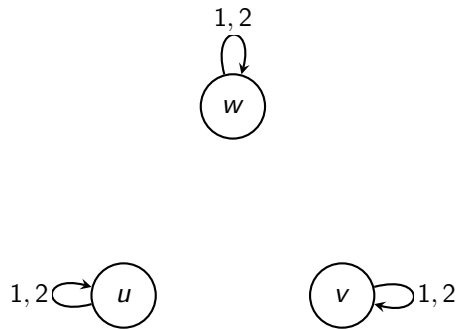
Realizing them as classical graphs

Each nontrivial class ($m = 2, 4, 6, 8$) is exhibited as the **derived quantum graph** of an explicit classical voltage graph over $\mathbb{Z}_3$.

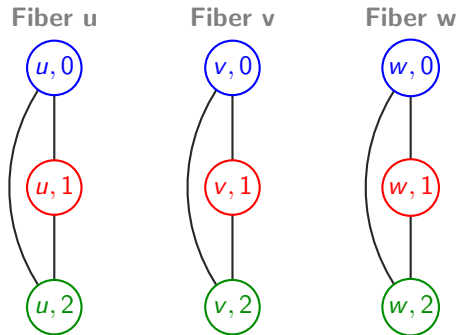
- $m = 2$: loops only
- $m = 4$: loops + oppositely-oriented triangles
- $m = 6$: loops + partial $K_{3,3}$ between fibers
- $m = 8$: full $K_{3,3}$ (complete graph $\mathfrak{sl}_3$)

By a theorem of Schäfer and Tobolski: each is **quantum isomorphic** to a classical graph on **9 vertices**.

Classical graphs quantum isomorphic to our graphs

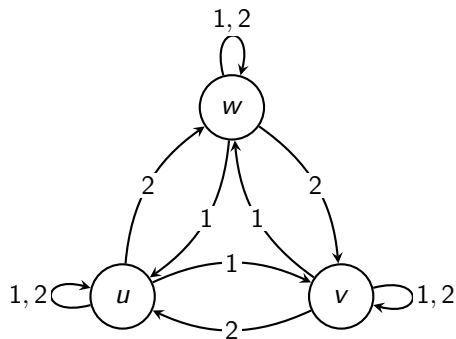


Base Γ : directed loops

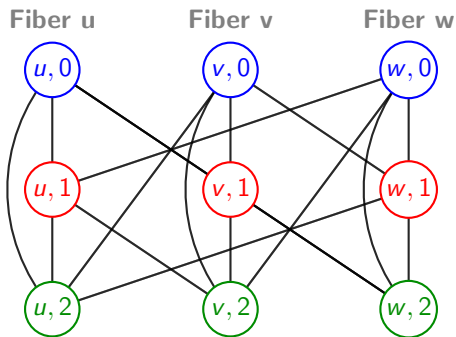


Derived Γ^λ : undirected 3-cycles

Figure: Voltage graph and derived cover for the loopfree, undirected quantum graph on $M_3(\mathbb{C})$ with $m = 2$ quantum edges.

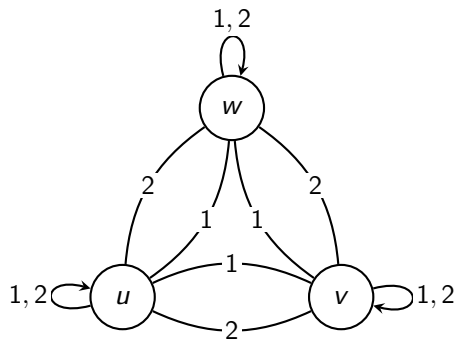


Base Γ : directed triangle & loops

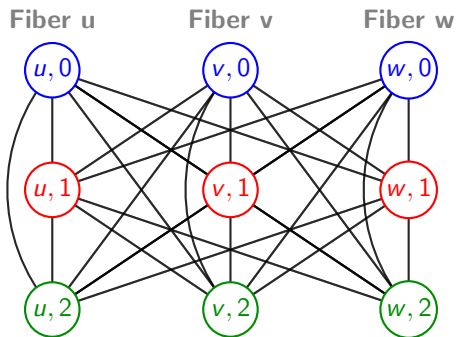


Derived Γ^λ

Figure: Voltage graph and derived cover for the loopfree, undirected quantum graph on $M_3(\mathbb{C})$ with $m = 4$ quantum edges.

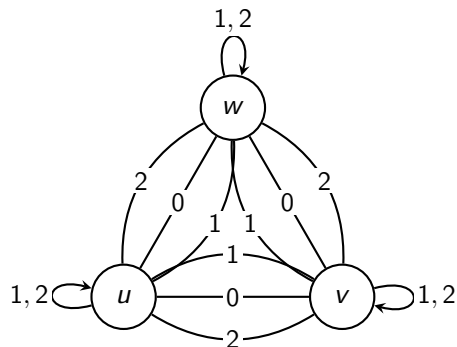


Base Γ : directed multigraph

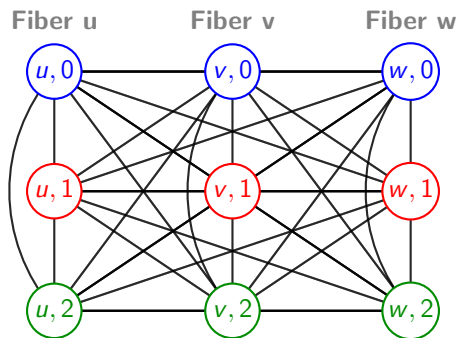


Derived Γ^λ : partial $K_{3,3}$ between fibers

Figure: Voltage graph and derived cover for the loopfree, undirected quantum graph on $M_3(\mathbb{C})$ with $m = 6$ quantum edges.



Base Γ : full directed multigraph



Derived Γ^λ : perfect $K_{3,3}$ between all fibers

Figure: Voltage graph and derived cover for the loopfree, undirected quantum graph on $M_3(\mathbb{C})$ with $m = 8$ quantum edges.

Vertex-transitivity

Vertex-transitive quantum graph: a quantum graph whose Gromada subspace V is such that $\text{span}(\{u \in U(n) : uVu^* = V\}) = M_3(\mathbb{C})$.

Proposition

Every self-reflective subspace generated by P, Q gives a **vertex-transitive** quantum graph.

- Proof idea: conjugation by any $P^a Q^b$ fixes each generator up to phase $\Rightarrow$ the 9-element group $\{P^a Q^b\}$ stabilizes *every* such subspace
- Holds already at minimal dimension $m = 2$

Comparison: Hayes et al. classify *all* vertex-transitive quantum graphs on $M_3(\mathbb{C})$ – a strictly larger family than ours.

Where the $M_2(\mathbb{C})$ story breaks down

$M_2(\mathbb{C})$: simple graphs classified purely by m (via $PU(2) \cong SO(3)$).

Lemma

$PU(3) \curvearrowright \mathbb{P}(\mathfrak{su}(3))$ has **infinitely many orbits**.

- Proved via a projective invariant $I(X) = c(X)^2/q(X)^3$
- Consequence: infinite families of mutually non-isomorphic undirected graphs on $M_3(\mathbb{C})$ for fixed m

Summary

- Classified P, Q -generated quantum graphs on $M_3(\mathbb{C})$: exactly **5** isomorphism classes
- Realized them as **classical** graphs on 9 vertices, via voltage/derived quantum graphs
- All are **vertex-transitive**
- Showed the clean $M_2(\mathbb{C})$ picture **fails** for $M_3(\mathbb{C})$: infinite families exist, and non-classical examples exist

Further work: extend to $M_n(\mathbb{C})$, $n \geq 4$ – no exceptional isomorphism like $PU(2) \cong SO(3)$ is available.

Thank you!